

Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
1. 41×8	$8(40 + 1)$ $= 320 + 8$ $= 328$	$\underline{8}$ $\underline{40}$ 320 + $\underline{1}$ 8	328
2. 6×89	$6(80 + 9)$ $= 480 + 54$ $= 534$	$\underline{6}$ $\underline{80}$ 480 + $\underline{9}$ 54	534
3. 96×3	$3(90 + 6)$ $= 270 + 18$ $= 288$	$\underline{3}$ $\underline{90}$ 270 + $\underline{6}$ 18	288

For each expression, generate an equivalent numeric expression using the distributive property and then find the value.

4. 9×42

$$\begin{aligned}
 &9(40 + 2) \\
 &= 9 \times 40 + 9 \times 2 \\
 &= 360 + 18 \\
 &= 378
 \end{aligned}$$

5. $28 \cdot 7$

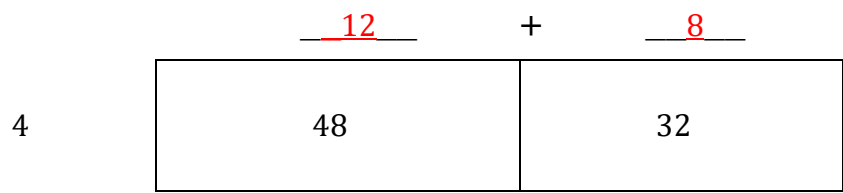
$$\begin{aligned}
 &7(20 + 8) \\
 &= 7 \times 20 + 7 \times 8 \\
 &= 140 + 56 \\
 &= 196
 \end{aligned}$$

6. 5×64

$$\begin{aligned}
 &5(60 + 4) \\
 &= 5 \times 60 + 5 \times 4 \\
 &= 300 + 20 \\
 &= 320
 \end{aligned}$$

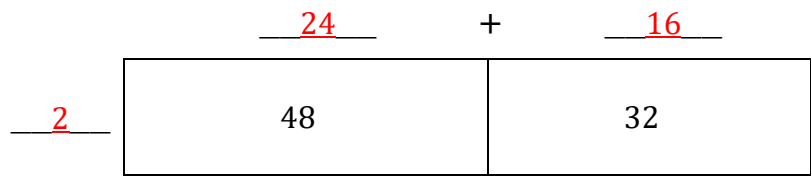
Consider the addition expression $48 + 32$.

7. Complete the area model to rewrite the expression using the distributive property.



$48 + 32 = 4 (12 + 8)$

8. Use a different common factor of 48 and 32 to complete the area model and rewrite the expression using the distributive property.
Answers may vary.



$48 + 32 = 2 (24 + 16)$

Complete the table by rewriting each addition expression using the distributive property, modeling with area, and finding the sum.

Addition Expression	Distributive Property	Area Model	Product
9. 60 + 165	$\begin{aligned} &15(4 + 11) \\ &= 15 \times 4 + 15 \times 11 \\ &= 60 + 165 \\ &= 225 \end{aligned}$	<u>4</u> 60 <u>11</u> 165 <u>15</u> +	225
10. 240 + 96	$\begin{aligned} &24(10 + 4) \\ &= 24 \times 10 + 24 \times 4 \\ &= 240 + 96 \\ &= 336 \end{aligned}$	<u>10</u> 240 <u>4</u> 96 <u>24</u> +	336